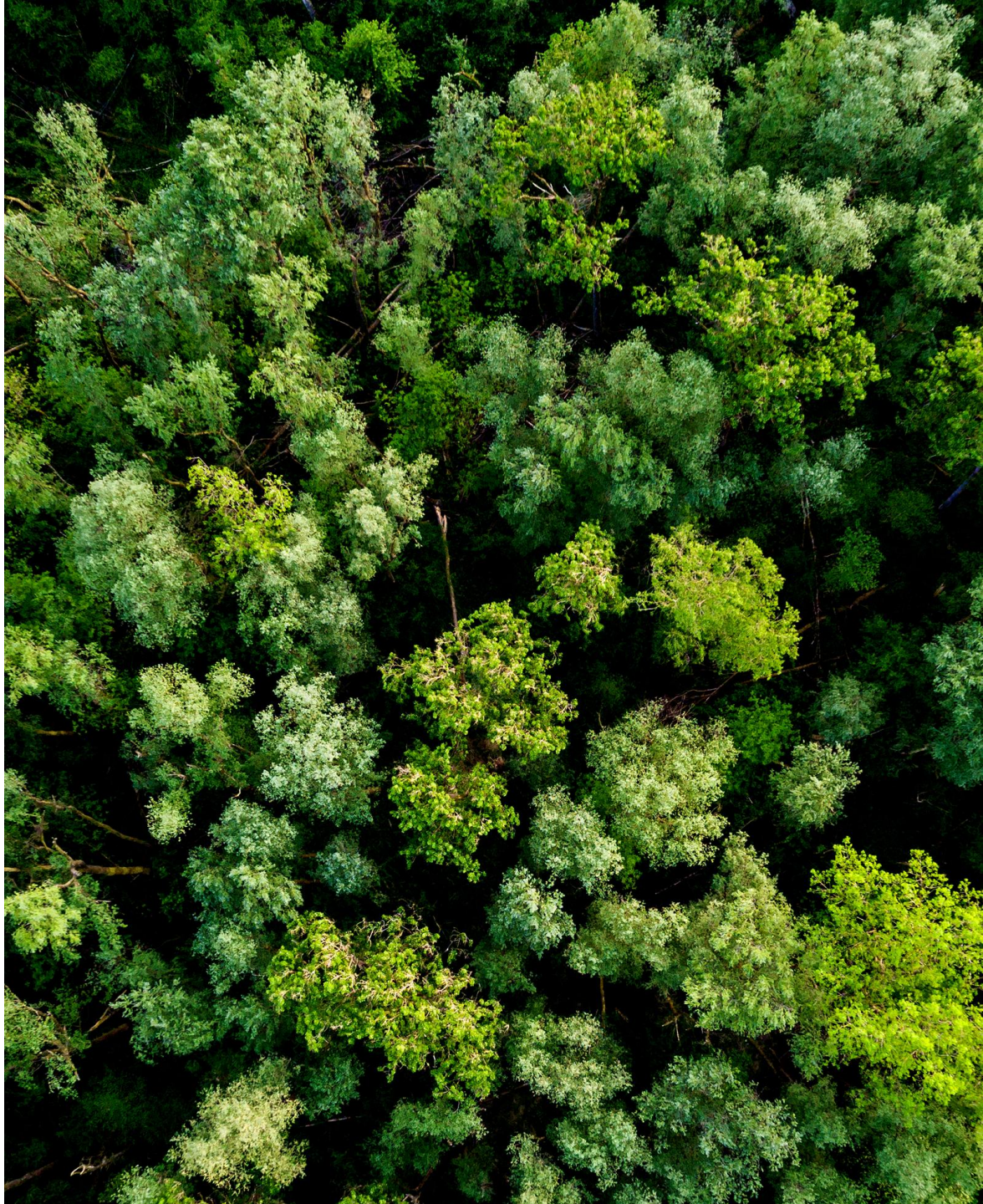
RANDOM FOREST

Use case: The CMS classification of colorectal cancer

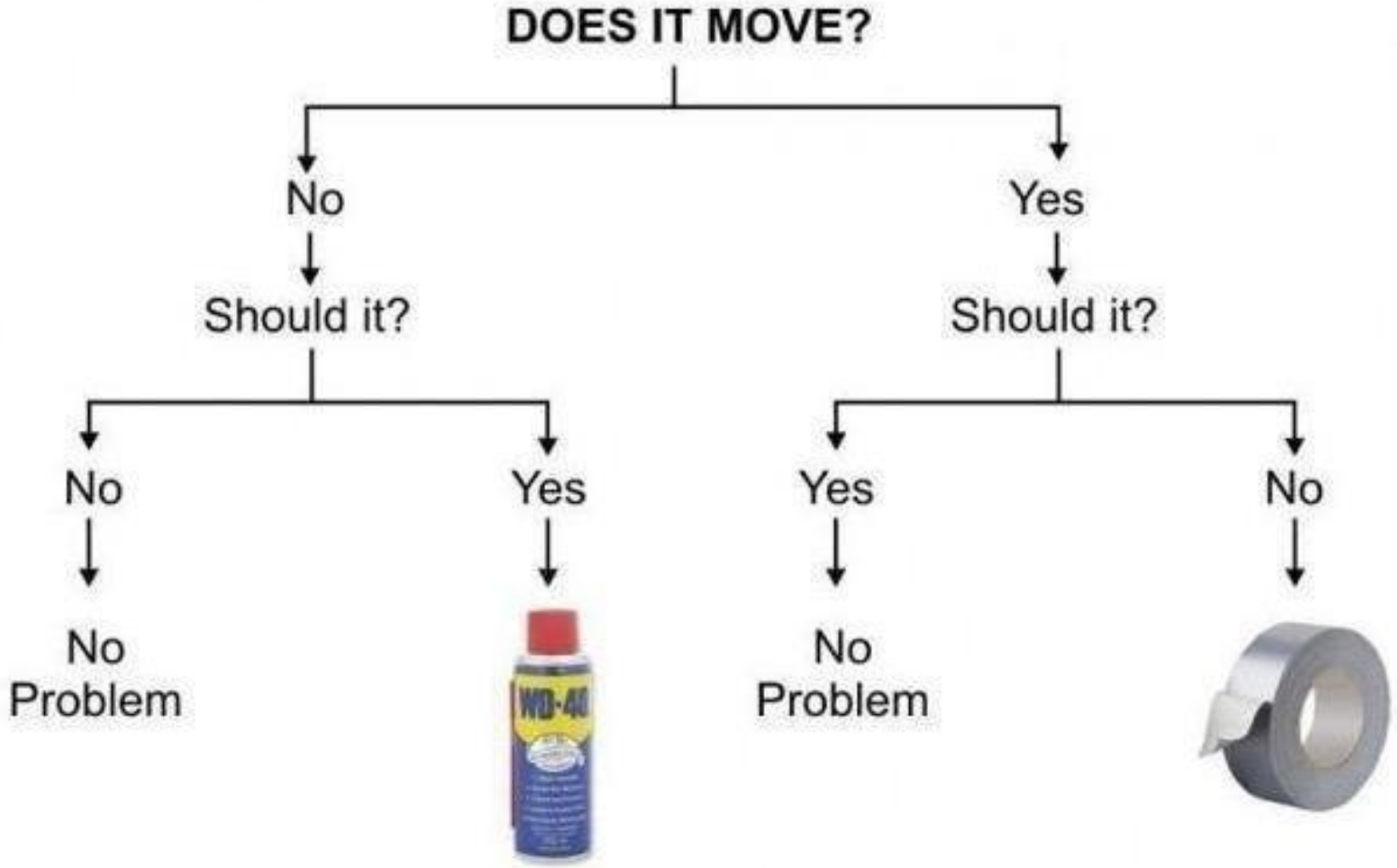
Inbar Leaf, ZHAW, 2025



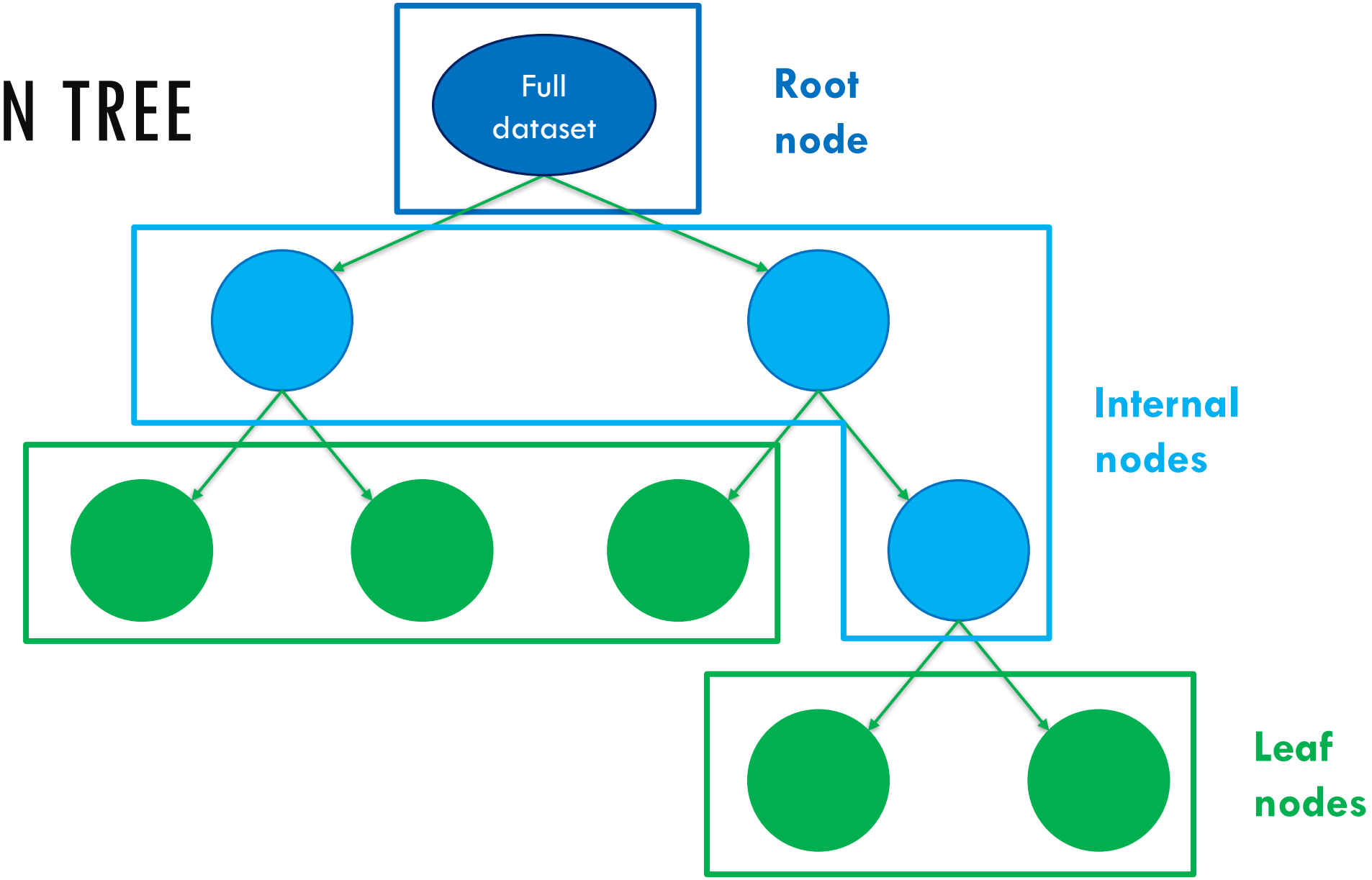
AGENDA

- Decision trees
- Random Forest
- The CMS classification of CRC
- Practice
- Project presentation

DECISION TREE

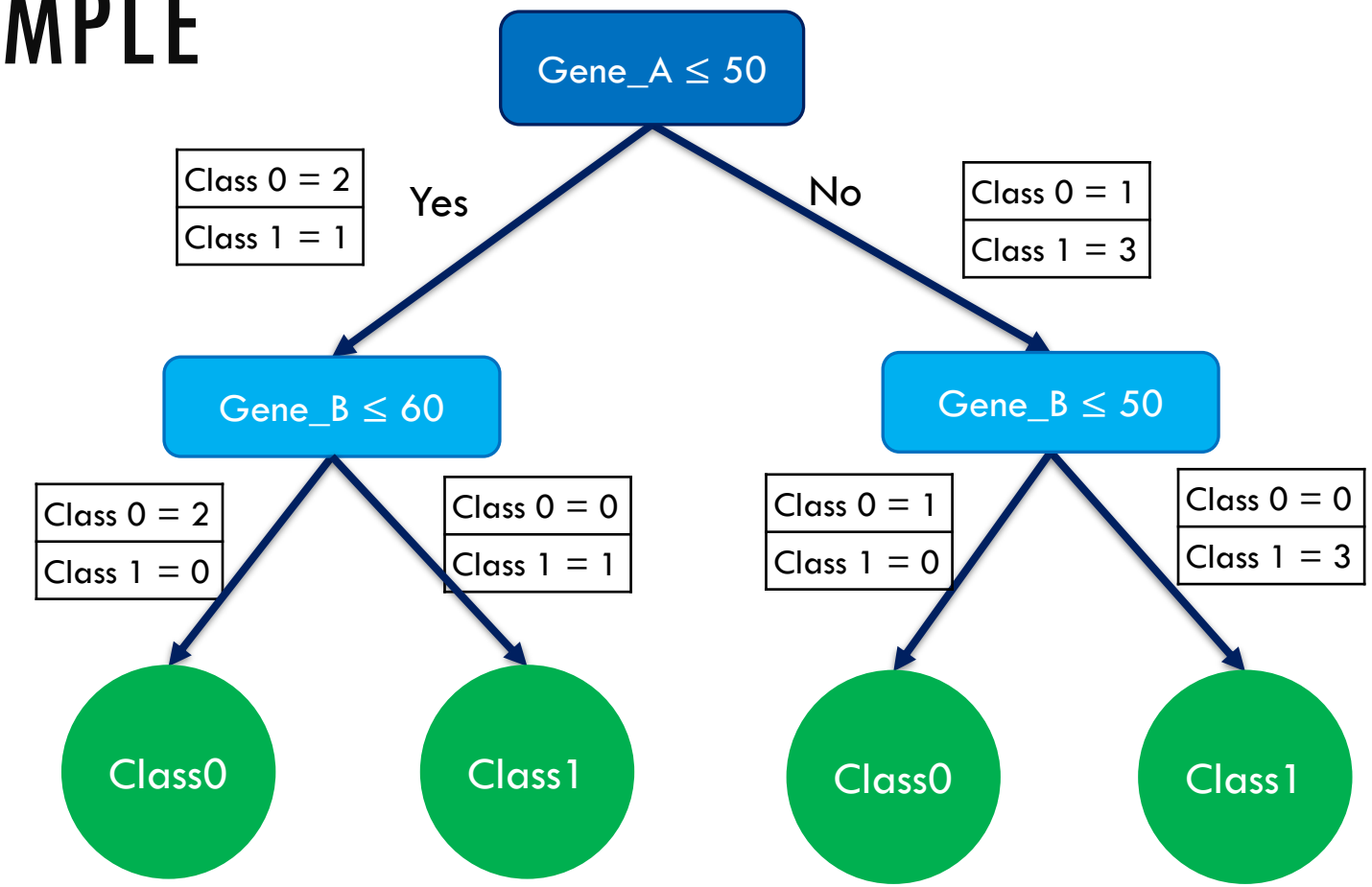


DECISION TREE



DECISION TREE: EXAMPLE

Sample	Gene_A	Gene_B	Class
1	20	30	**0**
2	30	55	**0**
3	70	25	**0**
4	60	80	**1**
5	80	90	**1**
6	25	95	**1**
7	75	70	**1**

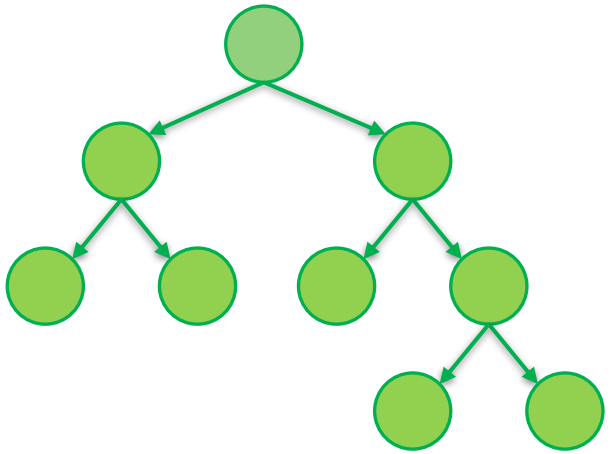


Splitting criteria:

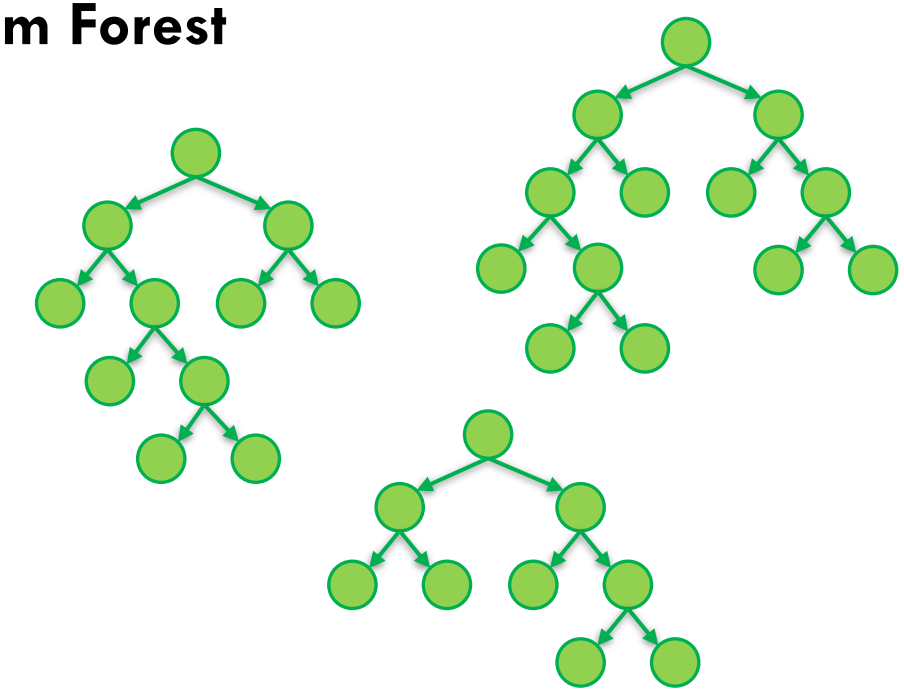
- Gini impurity
- Entropy

RANDOM FOREST

Decision tree

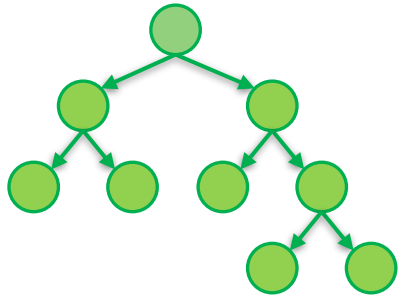


Random Forest

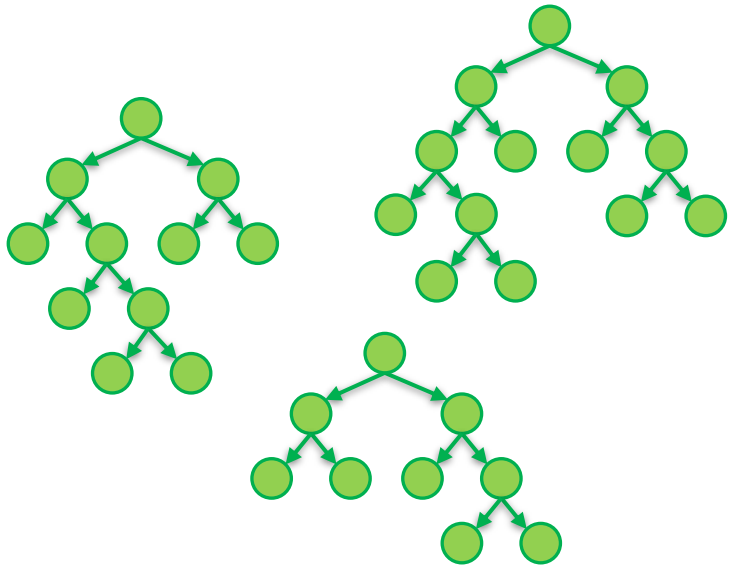


RANDOM FOREST

Decision tree



Random Forest



How does Random Forest work?

Bootstrapping

(sampling with replacement)

Build decision trees

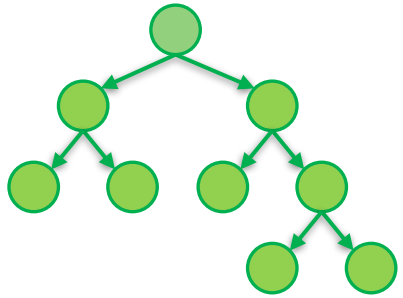
Select a random subset of the features at each split

Merge predictions

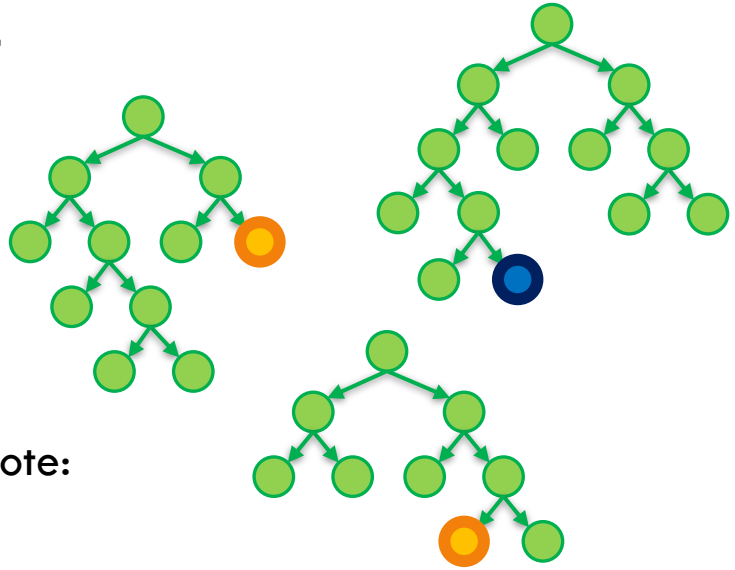
Classification: Majority votes	Regression: Average

RANDOM FOREST

Decision tree



Random Forest



Majority vote:
orange

How does Random Forest work?

Bootstrapping

(sampling with replacement)

Build decision trees

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RANDOM FOREST

Pros

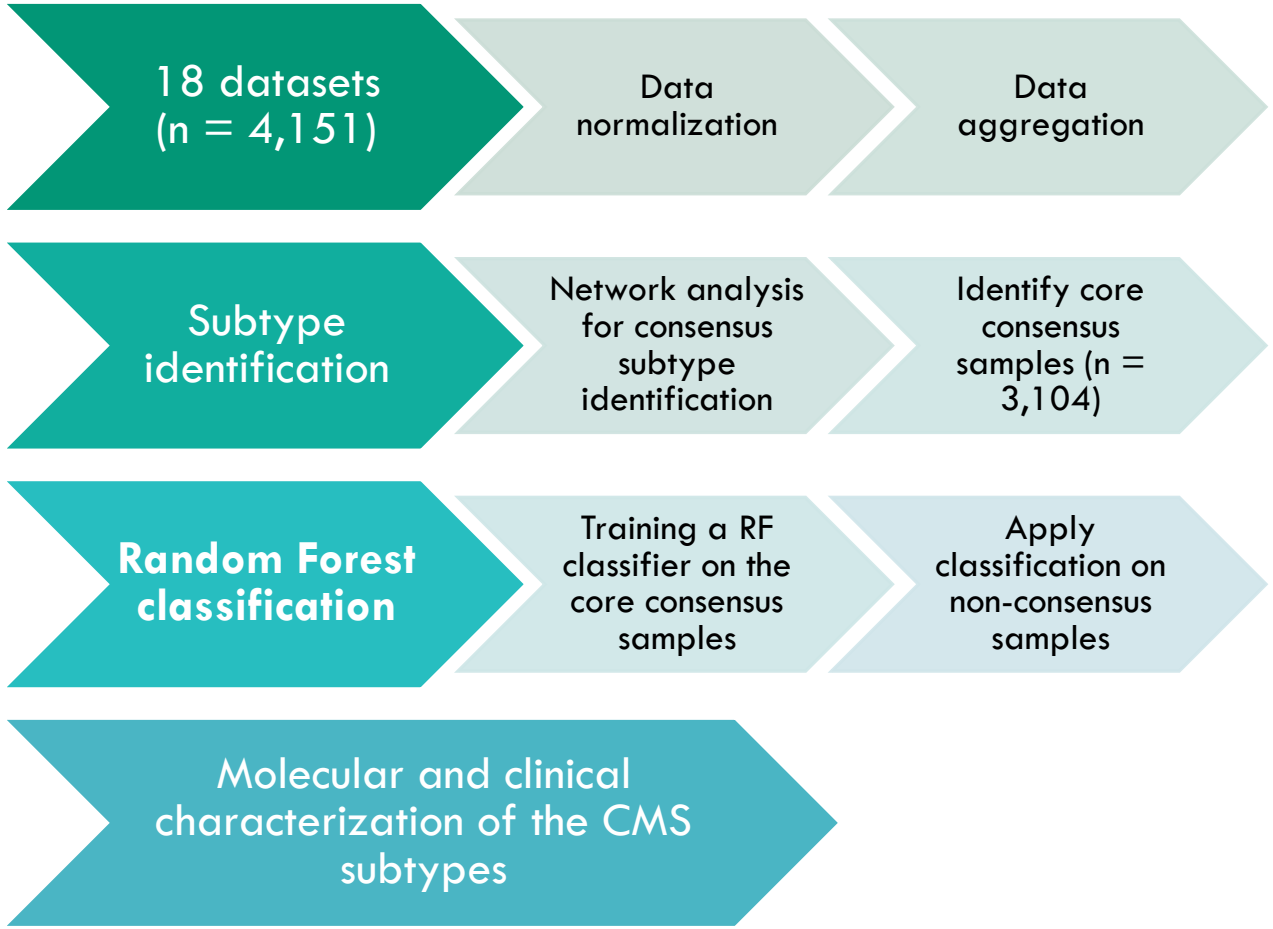
- Robustness to noise and outliers
- Less prone to overfitting
- Can handle high-dimensional data
- Non-parametric
- Provides feature importance scores

Cons

- Slower prediction time and higher memory usage with larger forests or larger datasets
- Lack of interpretability

The consensus molecular subtype classification of colorectal cancer

CMS1 MSI immune	CMS2 Canonical	CMS3 Metabolic	CMS4 Mesenchymal
14%	37%	13%	23%
MSI, CIMP high, hypermutation	SCNA high	Mixed MSI status, SCNA low, CIMP low	SCNA high
BRAF mutations		*KRAS* mutations	
Immune infiltration and activation	WNT and MYC activation	Metabolic deregulation	Stromal infiltration, TGF-β activation, angiogenesis
Worse survival after relapse			Worse relapse-free and overall survival



Guinney et al. 2015

PRACTICE

https://colab.research.google.com/drive/16pYM-MEdzpL4siaLFMdhcGVgq85ofTxY?usp=sharing

PROJECT: RANDOM FOREST CLASSIFICATION OF CMS: BINARY VS MULTI-CLASS APPROACHES

Goal:

- Apply RF classification with both muti-class and binary classification approaches, and compare them

Deliverables:

- Apply RF to CRC dataset labelled for CMS
- Compare binary and multi-class classification approaches
- Analyse and discuss the results, advantages/disadvantages, evaluation metric choice

Breiman, L. Random Forests. *Machine Learning* **45**, 5–32 (2001). https://doi.org/10.1023/A:1010933404324

Guinney, J., Dienstmann, R., Wang, X. *et al*. The consensus molecular subtypes of colorectal cancer. *Nat Med* **21**, 1350–1356 (2015). https://doi.org/10.1038/nm.3967

Alderdice, M., Richman, S. D., Gollins, S., Stewart, J. P., Hurt, C., Adams, R., McCorry, A. M., Roddy, A. C., Vimalachandran, D., Isella, C., Medico, E., Maughan, T., McArt, D. G., Lawler, M., & Dunne, P. D. (2018). Prospective patient stratification into robust cancer-cell intrinsic subtypes from colorectal cancer biopsies. *The Journal of pathology*, *245*(1), 19–28. https://doi.org/10.1002/path.5051

For further learning: How to Implement Random Forest From Scratch in Python: https://machinelearningmastery.com/implement-random-forest-scratch-python/